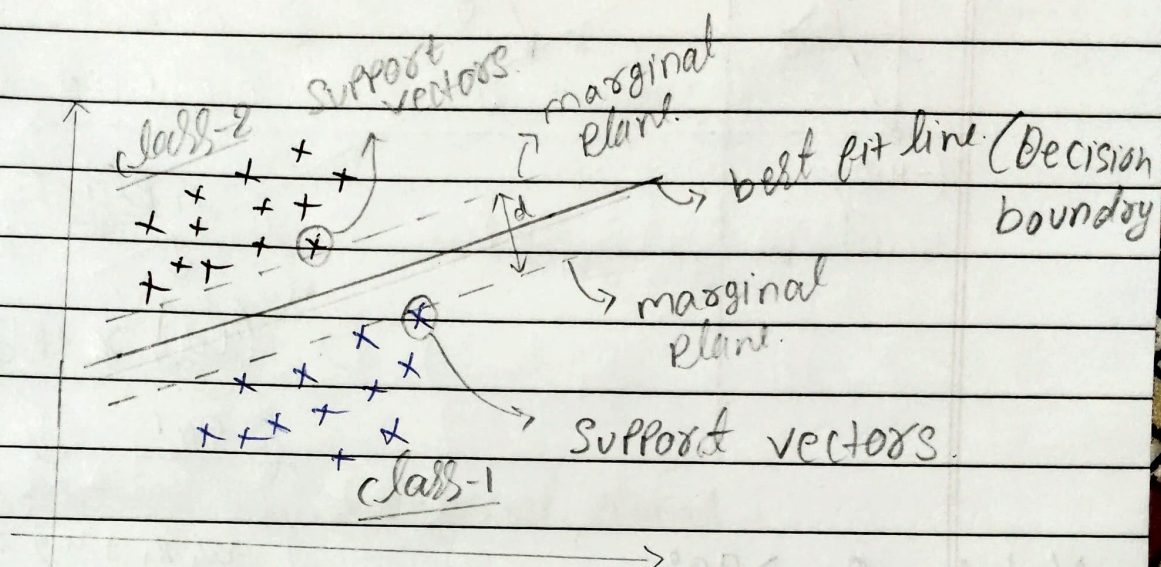
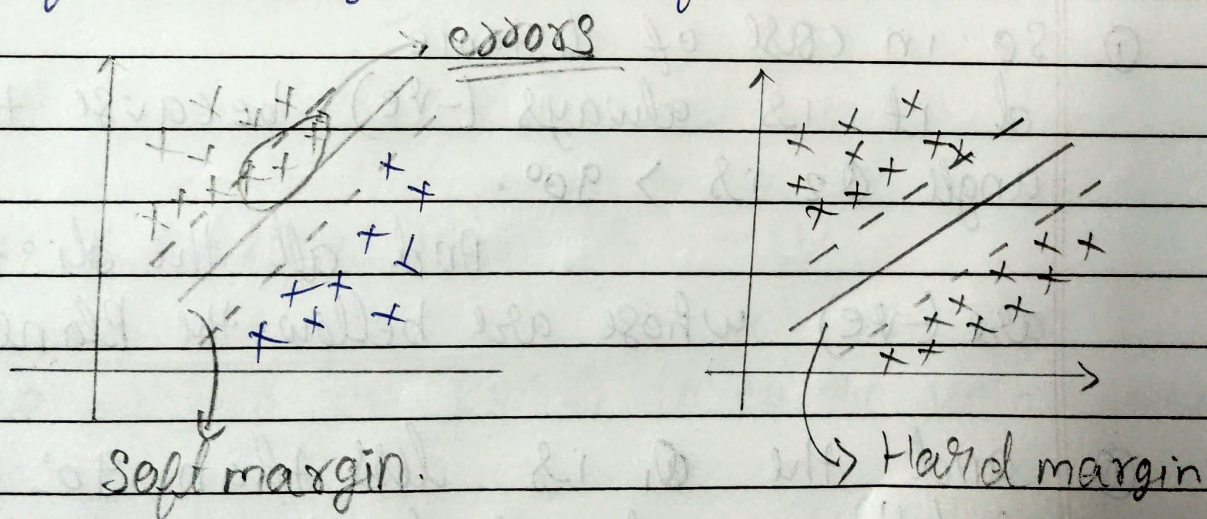


Support vector machine Algorithm.

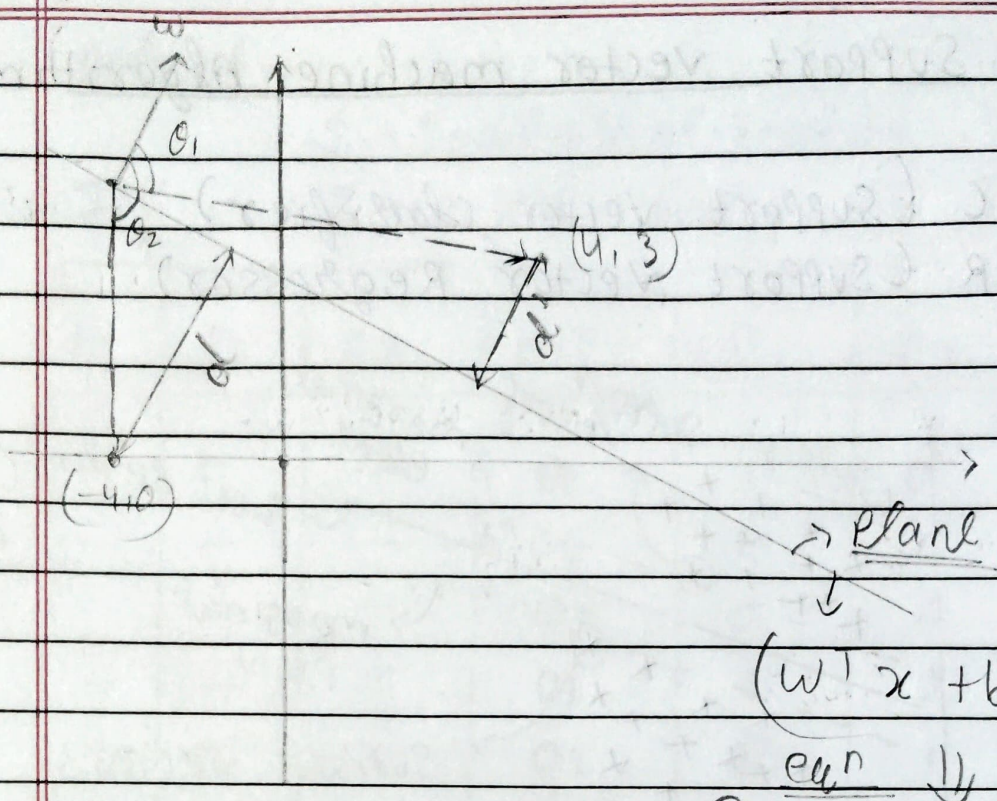
- (i) SVC (Support vector classifier)
- (ii) SVR (Support vector Regressor)



we try to keep the distance is maximum of the marginal planes from the decision boundary



if we get an error to fitting the line then it is called as soft margin.

Note:

$\theta_2 > 90^\circ$

$\theta_1 > 0^\circ$ and $\theta_1 < 90^\circ$

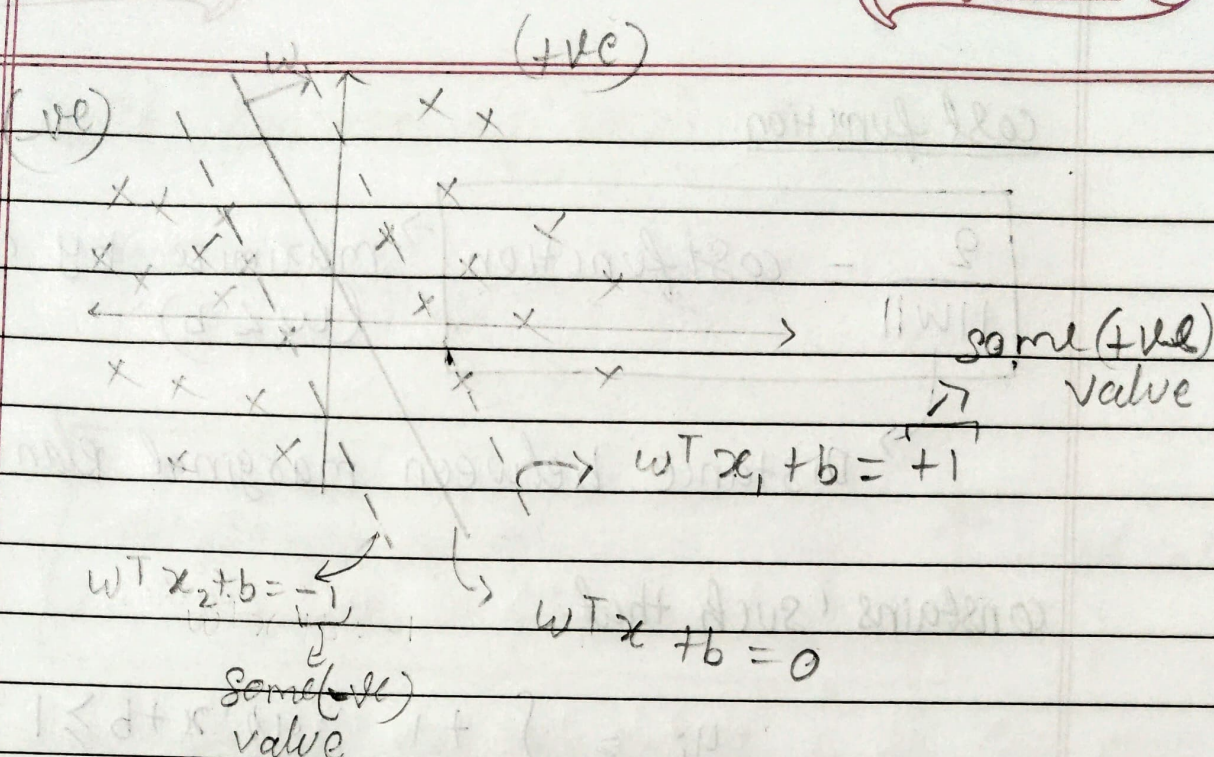
$w_1 x_1 + w_2 x_2 + b = 0$

$\Downarrow$
 $ax + by + c = 0$

- ① So in case of distance d it is always (-ve) because the angle θ_2 is $> 90^\circ$.

And all the distances are (-ve) whose are below the Plane.

- ② and the θ_1 is less than 90° so it (d') is always (+ve) and the distances whose are present above the Plane is (+ve).



⇒ Subtracting the marginal planes.

$$w^T x_1 + b = +1$$

$$\underline{w^T x_2 + b = -1}$$

$$w^T x_1 - w^T x_2 = +2$$

$$\Rightarrow w^T (x_1 - x_2) = +2 \quad \text{--- ①}$$

Converting the eqn ① into the unit vector

$$\frac{w^T (x_1 - x_2)}{\|w\|} = \frac{2}{\|w\|}$$

∴ $\|w\| \rightarrow$ magnitude of w

cost function.

$$\frac{2}{\|w\|} = \text{cost function} \rightarrow \text{maximize by changing } (w \text{ \& } b)$$

Distance between marginal planes.

constraints such that.

$$y_i = \begin{cases} +1 & ; w^T x + b \geq 1 \\ -1 & ; w^T x + b \leq -1 \end{cases}$$

$\Downarrow$

correctly classified
Points.

for all the correctly predicted point.

$$\rightarrow \text{Constraint} \Rightarrow y_i * (w^T x + b) \geq 1$$

Cost function of SVM (SVC)

$$\underset{(w, b)}{\text{maximize}} \quad \frac{2}{\|w\|} = \underset{(w, b)}{\text{minimize}} \quad \frac{\|w\|}{2}$$

$$\boxed{\text{cost function} = \underset{(w, b)}{\text{minimize}} \quad \frac{\|w\|}{2}} \rightarrow \text{Ideal case}$$

→ Real world scenario cases:

We have to add the Hyperparameters.

$$\underset{w, b}{\min} \quad \frac{\|w\|}{2} + C_j \sum_{i=1}^n \eta_i \rightarrow \text{eta}$$

⇓

soft margin

Hyperparameter

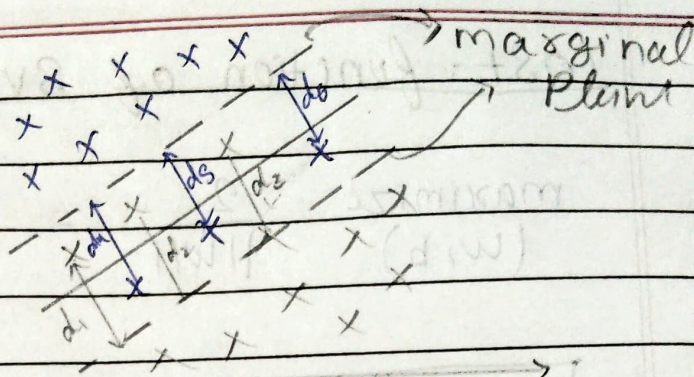
⇓

How many points we want to avoid misclassification.

→ Hinge Loss.

$$\sum_{i=1}^n \eta$$

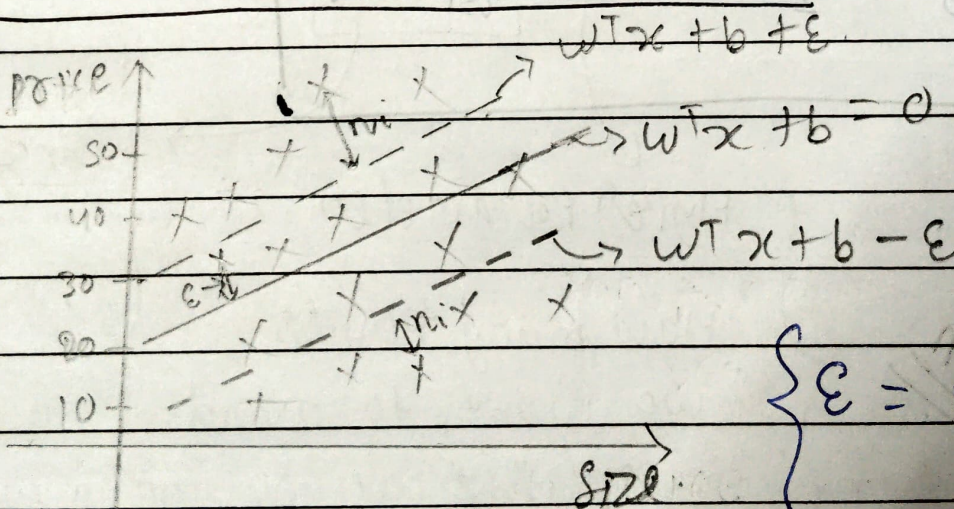
⇒ summation of the distance of the incorrect data points from the margin planes.



$C_i = 6$ (6 Points are miss classified.)

$$\sum_{i=1}^6 \eta_i = d_1 + d_2 + d_3 + d_4 + d_5 + d_6$$

Support Vector Regression:



$\left\{ \begin{array}{l} \epsilon = \text{marginal} \\ \text{error} \end{array} \right\}$

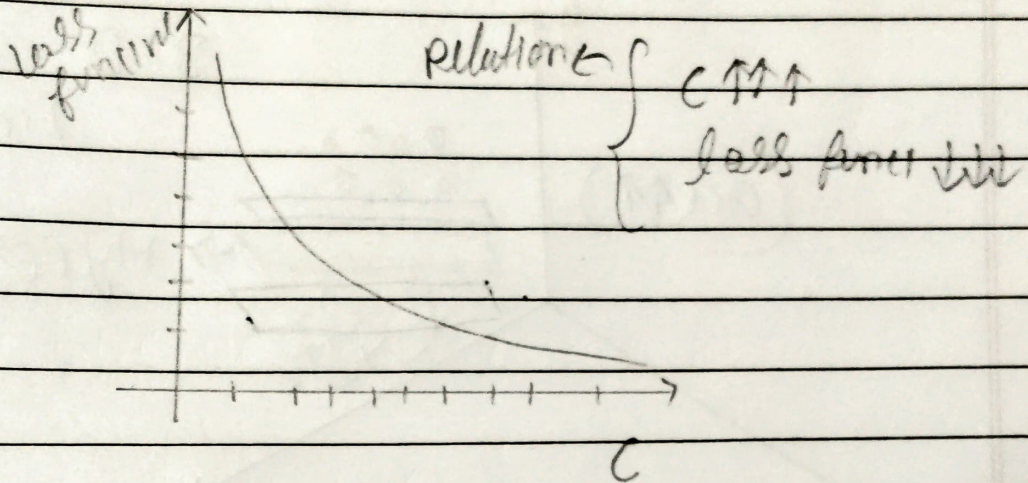
Cost function:-
$$\min_{w, b} \frac{\|w\|^2}{2} + C_i \sum_{i=1}^n \eta_i$$

$$\text{constrain} = |y_i - w_i x_i| \leq \epsilon + \eta_i$$

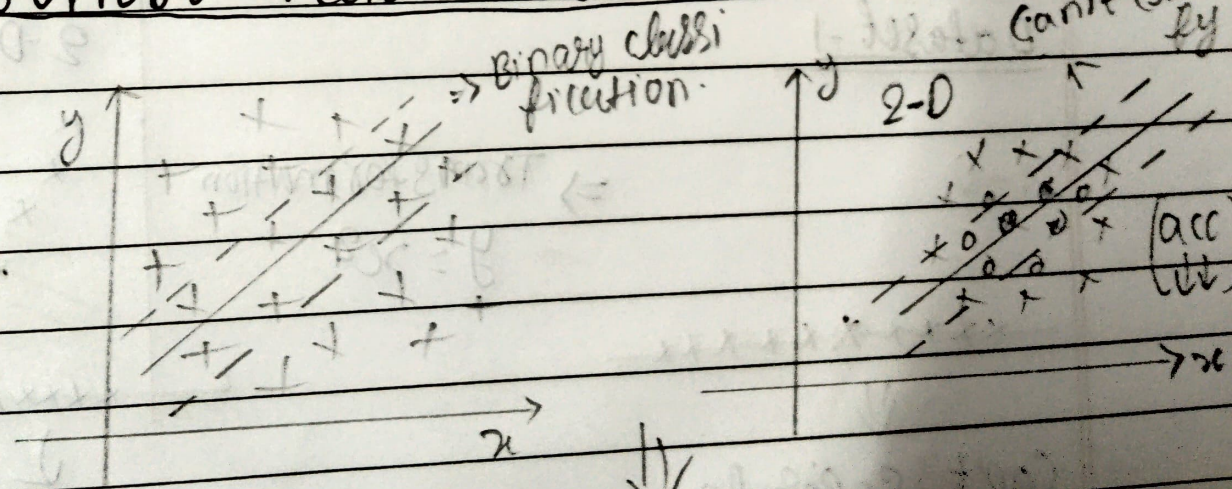
 $\Downarrow$

loss function.

here η_i is the distance between the point which is outside from the margin plane.



Support vector machine Kernels.


 $\Downarrow$

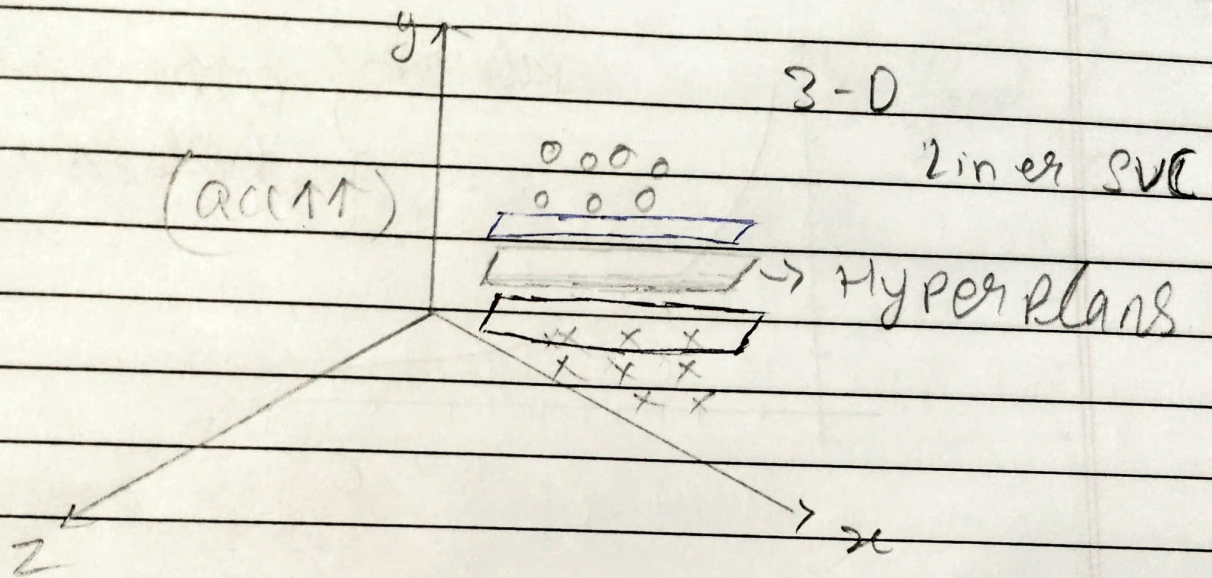
using SVM kernels.

we apply the transformations to



mathematical formula

Converting ~~2-D~~ 2-D into the 3-D



Example

Dataset - 1

1-D

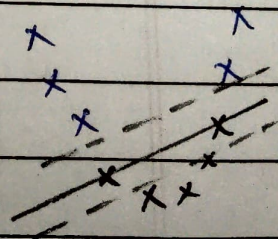
⇒ Transformation
 $y = x^2$

— x x x x x x x x —



Can't separate
with SVC

2-D



separable
with SVC